

STORAGE, MEMORY AND PROCESSING DEVICES

CHAPTER 3

CHAPTER OVERVIEW

Unit 3.1 Storage devices

Unit 3.2 Processing devices

Unit 3.3 Basic troubleshooting

By the end of this chapter, you will be able to:

- Describe the difference between primary and secondary storage
- Discuss online storage
- List the best storage media for backups and archiving
- Describe the role and function of various internal computing components, such as the motherboard, CPU, RAM and ROM
- Interpret advertisements for storage media and computers
- Do basic troubleshooting on input, output, storage and processing devices

INTRODUCTION

In this section, you will get more detail about the purpose of storage, what the difference is between primary and secondary storage, why computers need storage and what online storage is. You will be introduced to the storage devices that do the processing in a computer, such as the motherboard, random-access memory (RAM), read-only memory (ROM) and central processing unit (CPU) and what their functions are.

3.1 Storage devices



Something to know

One gigabyte (GB) is equal to 1 024 megabytes (MB), and one terabyte (TB) is equal to 1 024 GB.

Storage media have come such a long way since the early days of computing, and the size of data has grown so much that it is impossible to imagine that all it took to get humans to the moon was about 600 MB of data.

PRIMARY STORAGE

Primary storage is also called the main or internal memory of the computer. This storage is accessed directly by the CPU and is where the main instructions for the computer's operations are stored. The CPU can access these instructions and execute them as they are needed.

Primary storage is most often the RAM and the ROM, which will be discussed later in this chapter. The main function of the primary storage devices is to hold data for a short period of time while the computer is running and power is going to the devices. This data is removed once the power is stopped (like when the computer is shut down).

SECONDARY STORAGE

Secondary storage is also called external, secondary or auxiliary memory. This type of storage holds data for the long term. Data stored on secondary storage devices can only be removed by deleting it. Secondary memory is where the operating system, hardware drivers and data created by the user is kept and stored permanently.

This means that, in the case of power failures, secondary storage will preserve the information that is saved to it, even if the power to the computer is lost, while the data on the primary storage devices will be lost.

The most common example of secondary storage is the hard drive inside a computer. Other examples include solid-state drives (SSDs), USB flash drives and secure digital (SD) cards.

Table 3.1: Comparison: Memory versus storage

MEMORY	STORAGE
Memory needs power to keep its contents (volatile). The RAM in a computer is an example of volatile memory . This means that data is not stored permanently but is kept for short-term use only. The data is cleared as soon as the power to the RAM is stopped.	Keeps its contents even when the computer is switched off (non-volatile). Storage holds programs and data more permanently.
Memory is fast to access. It is more expensive per GB than storage.	Because magnetic and optical storage works mechanically, it is slow. Even electronic storage operates at lower speeds than memory.

STORAGE MEDIA FOR BACKUP

One purpose of storage devices is to back up or **archive** your important data. In the business world, there is a need to store data permanently and in a way that does not get destroyed, corrupted or damaged easily. Different types of storage media can be used for backing up or archiving.

Backups are records of important information that is used often. Backups are kept for a relatively short period of time. Table 3.1 lists the advantages and disadvantages of different types of storage media used for backups.

BACKUP DEVICES

Backup refers to the process of making copies of data or data files to use in the event the original data or data files are lost or destroyed or storing the copies in a different place from where you keep your computer.

The most popular backup media to use are:

- **External hard drives (HDD):** These are quite fast and generally reliable. However, if an external hard drive is used often, the mechanical stress can shorten its lifespan. External hard drives are fragile.
- **External SSDs:** SSDs have no moving parts. SSDs are more expensive than hard drives and the storage space is limited due to the cost (the more space you get, the more expensive the SSD becomes).
- **USB flash drives:** USB flash drives are cheap and relatively easy to use. There are no moving parts, so they are an excellent way to move data around. However, they can be slow and unreliable.
- **Tapes:** Tape backup is copying data from a primary storage device to a tape cartridge so the data can be recovered if there is a hard disk crash or failure. Tape backups can be done manually or be programmed to happen automatically with appropriate software.
- **CDs and DVDs:** CDs and DVDs are still occasionally used but writing a CD or DVD is much slower and less convenient than using other media. They have a lower capacity than most flash drives and much lower capacity than portable hard drives.
- **Online storage/backup services:** Online data storage is virtually unlimited in size. This means that there is plenty of space for you to back up your entire hard drive, including all of your purchased programs, music, photos, and files. Saving to the cloud means that you are saving to hard drives in a remote location. You must keep your username and password safe in order to protect your data because if someone knows (or even guesses) your credentials, it may result in loss of data.

Archiving, on the other hand, is the long-term storage of information that will be used in future. Archived data is data that is not actively used and is original data that has been removed from its original location. There are several different storage media that can be used for archiving.

Magnetic tape has a very large storage capacity (up to 180 TB), but can be destroyed very easily and the data can be lost.

Online storage is an option and online storage providers (such as Amazon AWS) have relatively inexpensive packages available for storage and archiving.

The data stored in a backup is a copy of the current and active operational data in use by a business. This includes files which are currently being accessed and changed on a regular basis. The files stored on an archive are generally no longer in use, not changing frequently and not required on a regular basis.



Activity 3.1

1. Multiple choice:
 - a. Which of the following cannot be used to back up files?

A. External HDD	B. USB drive
C. Optical disc	D. RAM
 - b. Which of the following is the most expensive storage?

A. RAM	B. HDD
C. SSD	D. iCloud
 - c. Which of the following is secondary storage?

A. RAM	B. CPU
C. HDD	D. ROM
2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1. Most often the most expensive equipment in a PC, e.g. processor.	A. Archiving
2.2. A storage device that is relatively rugged and virtually shockproof as it does not have moving parts.	B. 1 024 GB
2.3. A cheap storage device and backup device option.	C. Primary storage
2.4. The long-term storage of information that will be used in future.	D. Online storage
2.5. The size of one TB in GB.	E. USB
2.6. Also called external, secondary or auxiliary memory.	F. SSD
2.7. Very common and relatively cheap storage device.	G. HDD
2.8. Amazon AWS and Google Drive, for example.	H. Secondary storage

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. Primary storage is accessed by the motherboard.
 - b. Magnetic tape has a very large storage capacity but can be easily destroyed and the data can be lost.
 - c. Online storage has limited space and you must pay to get more space.
 - d. The main purpose of primary storage is to store data for a long period of time.
 - e. Archiving is the short-term storage of information.
4. Answer the following:
 - a. What is the main purpose of primary storage?
 - b. Briefly describe the difference between primary and secondary storage.
 - c. What is the difference between backing up and archiving data?
 - d. What storage devices would you recommend to someone who needs to store data on a server and someone who wants to store copies of their digital photographs? Motivate your answer.

3.2 Processing devices

Processing is the second stage of the information processing cycle you learned about in Chapter 1. Processing mainly uses the system components of the computer, such as the motherboard, CPU and RAM, for its tasks. In this unit, you will learn more about the role of each component of a computer system, how they work and what they do.

MOTHERBOARD

The motherboard is one of the most essential parts of a computer. It holds many of the components that allow a computer to function, such as the CPU, RAM and the connectors for input and output devices.



Figure 3.1: Computer motherboard with components

The motherboard is made of a thin piece of rigid, **non-conductive** material. A thin layer of copper or aluminium foil is printed onto the sheet. These circuits are called traces. They are very narrow and form the connections between the various components housed in the motherboard.

The role of the motherboard is to:

- Provide a place for other devices or interfaces to be installed (such as more memory or another **graphics card**).
- Distribute power to the various components.
- Act as a communication hub as the components send and receive information through the motherboard.

CENTRAL PROCESSING UNIT (CPU)

The CPU, which is also called the processor, carries out the instructions of a computer program. It is, basically, the “brain” of the computer and does the **arithmetical**, logical and input/output operations of a computer program. The speed of the CPU tells you how much data it can process in a specific time. The speed is measured in **gigahertz** (GHz).



Figure 3.2: Intel Core i7 CPU

HOW A CPU WORKS



https://www.youtube.com/watch?v=cNN_tTXABUA

All instructions, whether they have been generated by the system or the user, go through the CPU. This could be anything from the operating system executing a task to the user typing letters in a document.

WHAT ARE CORES IN A CPU?

A core is usually the basic computing unit of the CPU. A single core can run a single program, problem or context. A CPU may have one or more cores to perform tasks at any single time. Original CPUs had single cores, but manufacturers began adding more cores to increase the performance of the CPU. A CPU with two cores can split the task into two tasks and thereby complete it faster. This speeds up the computer because it is, essentially, able to multitask.

A dual-core processor has two cores, so it looks like two CPUs to the operating system. A **quad-core** CPU has four cores, an **octa-core** CPU has eight cores, and so on. These cores also make sure that the CPU is still small enough to fit into a single socket, taking up less space on the motherboard.

RANDOM-ACCESS MEMORY (RAM) AND READ-ONLY MEMORY (ROM)

While **RAM** and **ROM** are both types of memory, they perform different functions. Table 3.2 compares and contrasts the roles and functions of RAM and ROM.

Table 3.2: Roles and functions of RAM and ROM

	RAM	ROM
Role	RAM is also called random-access memory and is a specific type of data storage. It allows data to be accessed in random order, which means specific information can be found quickly.	The ROM, or read-only memory, stores the basic instructions for what needs to happen when the computer is switched on.
Function	Ram is where the CPU goes to fetch the instructions and data that it needs to work on. (RAM gets its contents by loading them from storage.) RAM is also temporary storage for data for programs that are running, allowing the programs to work faster.	It contains the basic code to get the computer started. The ROM is normally stored on the BIOS chip in the motherboard. The information stored in the ROM is often difficult to change and the data is not lost when the computer is powered down.
Image		

INTERPRET COMPUTER ADVERTISEMENTS

When you want to buy a new computer or storage device, there are several key factors to consider. By using what you know about these components, you will be able to interpret the information in advertisements for computers.



8 GB
RAM

1 TB
HDD



**39 cm (15.6") 15-Series
Intel Core i5 Laptop Bundle**

- Intel Core i5 - 7200U (Up to 3.10 GHz)
- Ethernet/Wi-Fi/Bluetooth
- Windows 10 Home
- Includes backpack and mouse
- 2-year collect-and-repair warranty (356609)

SAVE 1000

8999
PER BUNDLE

+ FREE



Figure 3.3: A computer advertisement

Using what you have learned so far in this chapter, you will be able to interpret any advertisement by looking at the following key pieces of information:

- How fast is the CPU in the computer? Remember that the faster the CPU works, the faster your computer will be able to process data. So, a 3 GHz processor is twice as fast as a 1,5 GHz one. In this example, the CPU speed is up to 3.10 GHz.
- How big is the memory? The more RAM you have, the more programs and applications you can run at once. Most computers come with 4 GB of RAM, which is good for users who do not need to do much more than check emails or create some documents. Users who need their computer to do slightly more can look at getting 8 GB of RAM, while 16 GB is good for those who will often be using more memory-intensive programs on their computers (such as games and videos and photo-editing software). In this example, the memory is 8 GB RAM.
- How much storage capacity does it have? Most computers come with a fairly large hard drive (about 500 GB). You can also buy external hard drives that can store terabytes of data. Make sure you get one that will suit your needs. In this example, the storage capacity is 1 TB.
- Which operating system does the computer use? The latest operating system will be more compatible with new programs, and will be more secure and able to support the drivers for your hardware. In this example, Windows 10 Home is used.



Activity 3.2

- Multiple choice:
 - The start-up process is stored on which hardware component?
 - BIOS
 - RAM
 - ROM
 - CPU
 - The speed of a CPU is measured in?
 - GHz
 - MHz
 - Hz
 - Calculations per second
 - The various components of a motherboard are connected by which metal?
 - Sulphur
 - Copper
 - Silica
 - Silver
- Match Column B with Column A:

COLUMN A	COLUMN B
2.1. This unit performs all the basic mathematical tasks and logical functions of the CPU.	A. Control unit
2.2. Memory that is lost when the device shuts down.	B. Volatile memory
2.3. The high-speed memory where instructions can be copied to and retrieved for the CPU.	C. ALU
2.4. This part of the CPU manages the components by reading and interpreting instructions from memory and turns them into signals.	D. CPU
2.5. The most essential part of the computer.	E. Cache
2.6. This unit carries out the instructions of a computer program.	F. Motherboard
2.7. This unit allows data to be accessed in random order.	G. ROM
2.8. Data on this unit is often difficult to change.	H. RAM

... continued



Activity 3.2

... continued

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. The motherboard distributes power to the computer.
 - b. The RAM is stored on the BIOS chip.
 - c. The motherboard is the brain of the computing operation.
 - d. Temporary files are stored on the CPU.
 - e. Data stored on the RAM is permanent.
4. Answer the following:
 - a. Briefly describe the role and function of the motherboard and CPU.
 - b. Explain RAM.
 - c. Describe ROM.
 - d. Which questions should you ask yourself when looking at an advertisement for a computer?

3.3 Basic troubleshooting

It is important for you to know how to do basic troubleshooting on your storage devices, especially if those devices contain very important information. This unit will look at some basic troubleshooting for storage media.

STORAGE AND PROCESSING DEVICES

Most of the basic troubleshooting guidelines for external storage devices can apply to internal storage and processing devices as well, although there are some specific scenarios where advice for storage media will not solve problems with RAM or ROM modules.

FORMATTING

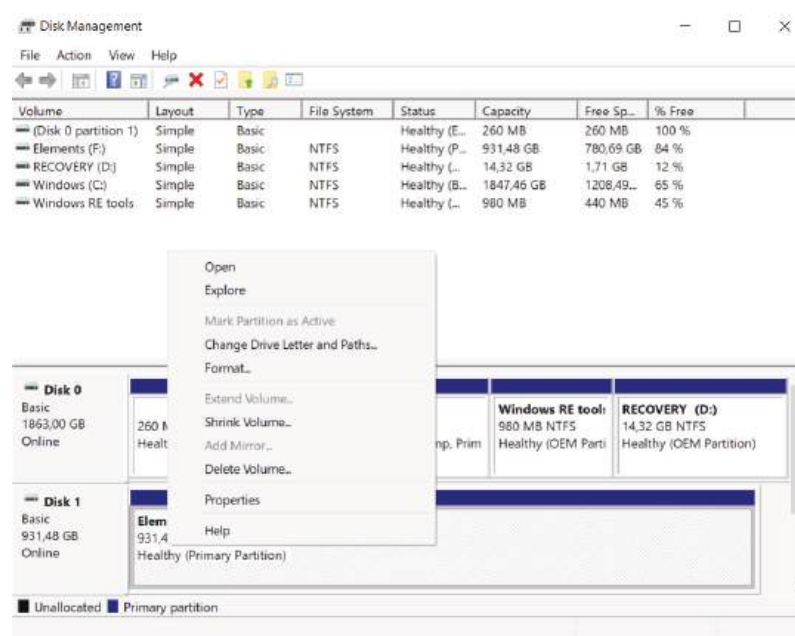
Formatting means preparing a disk to store data. Some disks, such as flash disks or hard drives, may need to be reformatted from time to time. Formatting will erase any data that is on the disk, so you will need to back up the data up before you **format**.

Windows 10 has a built-in disk management tool to make formatting a disk easier.

FORMATTING A DISK

To format a disk, you must do the following:

- Step 1:** Click on the *Search* icon on the bottom left corner of your screen.
- Step 2:** Type in *format*.
- Step 3:** Select *Create and format hard drive partitions*.
- Step 4:** This opens the *Disk Management* dialogue box.
- Step 5:** Right-click on the disk you want to format and click on *Format*.
- Step 6:** Choose *NTFS* (new technology file system) as the file system and select *Default* as the *Allocation Unit Size*.
- Step 7:** Click *OK* to format the drive. **BE VERY CAREFUL AS THIS ACTION WILL DELETE ALL DATA ON THE DRIVE.**



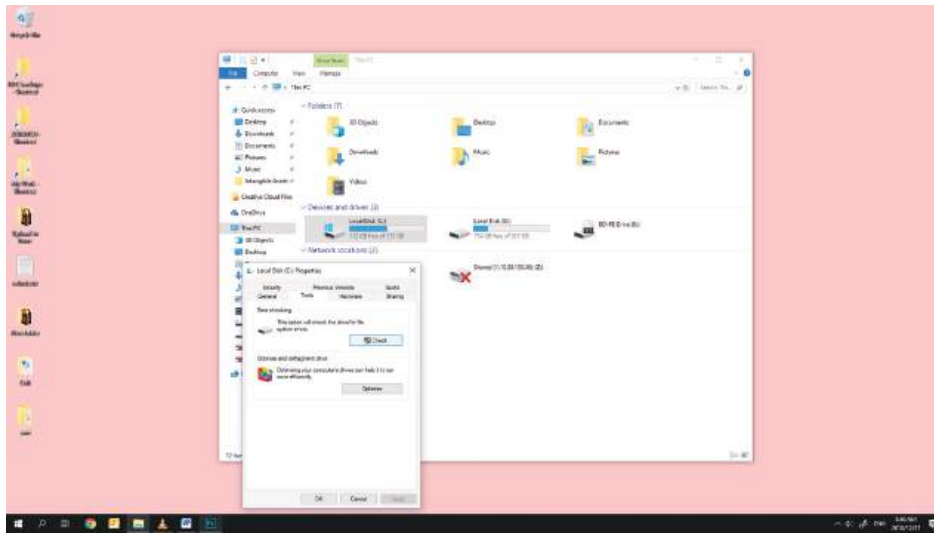
DISK SCANS

Disk scans can help detect and fix errors in USB flash drives, hard drives and SSDs. There is a disk scanner built into Windows 10.

USING A DISK SCANNER

To use the disk scanner, do the following:

- Step 1:** Click on the *Search* icon on the bottom left corner of your screen.
- Step 2:** Type in *Explorer* to open Windows Explorer (or press the Windows key + E).
- Step 3:** In the *File Explorer* window, click on *This PC* in the left navigation pane.
- Step 4:** A list of drives will be shown in the right window. Right-click on the drive you want to scan and select *Properties*.
- Step 5:** In the *Properties* window, click the *Tools* tab.
- Step 6:** Click the *Check* button in the *Error Checking* section.
- Step 7:** Run the scan without any interruptions.
- Step 8:** You may need to restart your computer if the scan detects any problems.



DISK DEFRAGMENTATION

Over time, the files on your computer's hard drive become scattered, making your computer slower. This is called **disk fragmentation**. **Defragmenting** your hard drive pulls these scattered bits of data back together. Windows 10 automatically defragments your hard drive once a week.

The process of defragmentation moves the data blocks on the hard drive around to bring all the parts of a file together. Defragmentation reduces file system fragmentation, increasing the efficiency of data retrieval and thereby improving the overall performance of the computer. At the same time, it cleans the storage and provides additional storage capacity.

DISK CLEANUP

While you are using your computer, you generate many types of files. These files take up space on your computer, making it slower. In Grade 10, you learned how to zip and unzip files, but this may not be enough to speed up your system.

Windows 10 also has a built-in disk cleanup tool that will delete the following:

- Temporary files from the internet.
- Deleted files and folders in the recycle bin.
- Temporary files created by the operating system.
- The component of the operating system you are not using.
- Applications or programs that you do not use.

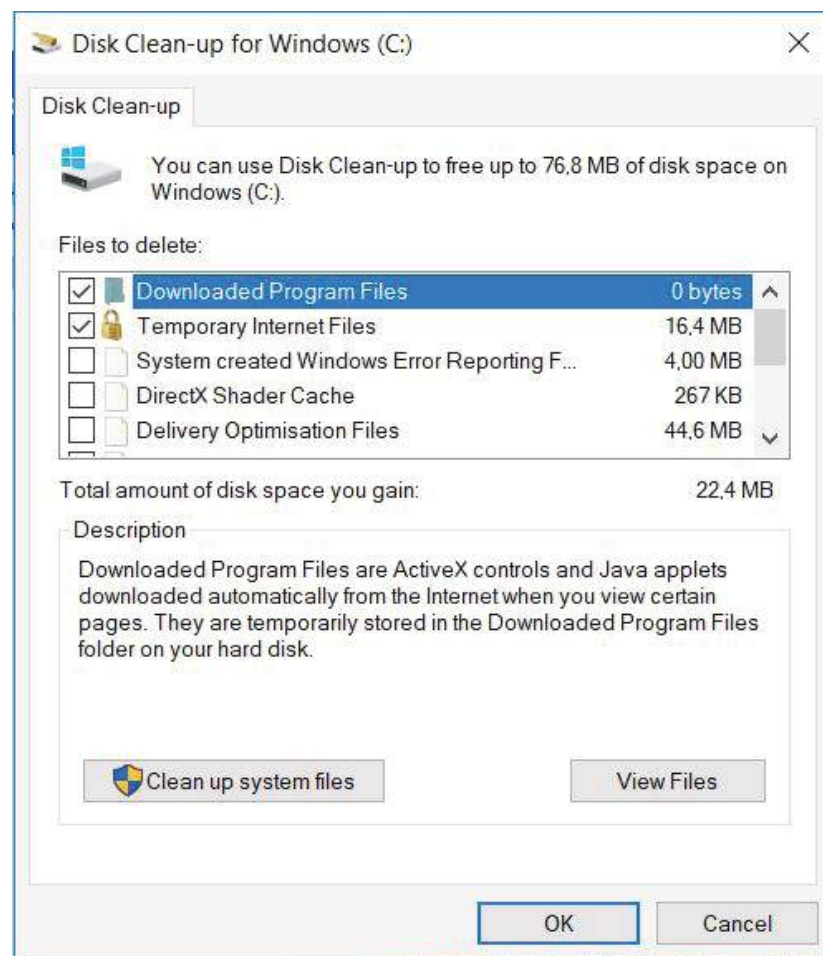


Figure 3.4: Disk cleanup helps with freeing up space on your computer



Activity 3.3

1. Match Column B with Column A:

COLUMN A	COLUMN B
1.1. Preparing a disk to store data.	A. NTFS
1.2. When the files on a disk are divided into pieces scattered around the disk.	B. Formatting
1.3. The process of locating segments of fragments of data stored on a hard drive and rearranging and restoring them into fewer fragments or into the whole file.	C. Disk fragmentation
1.4. A file system for a USB drive	D. Disk defragmentation
1.5. Window key + E is a shortcut to open.	E. Disk cleanup
1.6. A program that deletes temporary files	F. Windows Explorer
1.7. The window where you find the Tools tab when doing a disk scan.	G. The manufacturer's website
1.8. A resource you can use to find troubleshooting tips for output devices.	H. Properties

2. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
- Microsoft Windows 10 needs disk scanning software.
 - Formatting your device deletes program files.
 - You must make sure Num Lock is on if your Page Up/Page Down keys are not working.
3. Answer the following questions in your own words:
- Describe how you would format a disk.
 - What is the purpose of a disk scan?
 - Explain how you would manually defragment your disks.
 - What does the Windows 10 disk cleanup tool do?

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is an example of primary storage? (1)
- Motherboard
 - ROM
 - CPU
 - USB
- 1.2 Which of the following types of storage is volatile? (1)
- ROM
 - Internal HDD
 - RAM
 - Internal SSD
- 1.3 Which of the following is involved in the start-up of a computer? (1)
- CPU, ROM
 - RAM, ROM
 - RAM, CPU
 - HDD, ROM

... continued

REVISION ACTIVITY

... continued

- 1.4** When you look at a computer advertisement, what are the most important things to look at? (1)
- Storage, operating system, brand
 - Memory, processing speed, external hardware
 - Software, operating system, memory
 - Processing speed, storage, memory
- 1.5** Formatting means _____ a disk to store data. (1)
- Priming
 - Preparing
 - Saving
 - Processing

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- Data that is being processed is stored in memory. (1)
- The more cores a CPU has, the slower it operates. (1)
- RAM is related to storage, while the CPU is related to speed. (1)
- Before you format an SD card you must erase the information on it. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1-M). (6)

COLUMN A	COLUMN B
1. High-speed memory that can be used to copy or retrieve instructions.	A. Secondary storage
2. The start-up process of a computer.	B. Motherboard
3. A component that performs basic mathematical tasks and logical functions.	C. ALU
4. A device that physically connects all the internal components of a computer.	D. Control unit
5. A storage method that makes use of the internet.	E. Cache
6. The process of pulling scattered bits of data together.	F. Memory card
	G. Boot
	H. Disk defragmentation
	I. Online storage

QUESTION 4: FILL IN THE MISSING WORD(S)

Fill in the missing word(s) in the following statements. Provide only one word for each space.

- If you want to delete all the temporary files created by the operating system use i) _____. (1)
- Components on a motherboard are connected with ii) _____. (1)
- The speed of the iii) _____, which is measured in GHz, determines how much data your computer can process in a specific time. (1)
- The CPU fetches instructions from the hardware that need to be iv) _____. Then the CPU translates the instructions to v) _____. (2)

... continued

REVISION ACTIVITY

... continued

QUESTION 5: MEDIUM QUESTIONS

Look closely at the advert below and answer the following questions.

	Product Specifications Windows 10 Home 64 Intel Core i3-5005U 2.8 GHz 1 TB 5400 rpm SSD 4 GB DDR4-2133 SDRAM 802.11b/g/n and Bluetooth 4.0 combo 15.6" diagonal HD SVA BrightView WLED backlit (1366 × 768)
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- 5.1 What is the speed of the CPU in the computer? (1)
- 5.2 How big is the volatile memory? (1)
- 5.3 How much storage capacity does the computer have? (1)
- 5.4 Which operating system does the computer use? (1)
- 5.5 What does the 15.6" specification in the advert refer to? (1)
- 5.6 What does the 1366 × 768 specification refer to? (1)
- 5.7 What does the SSD acronym stand for? (1)
- 5.8 Give TWO reasons why users would prefer an SSD instead of an HDD? (2)
- 5.9 Suggest ONE way you can use to transfer data from this computer. (1)

QUESTION 6: SCENARIO-BASED QUESTIONS

Ronaldo has recently bought a new external hard drive for his laptop computer, since his laptop has started running out of space. Ronaldo's laptop has an SSD with a storage capacity of 500 MB, while his new external hard drive has a storage capacity of 1 GB.

- 6.1 Which of Ronaldo's storage devices has the longest lifespan? Give one reason for your answer. (2)
- 6.2 When Ronaldo connects his external hard drive to his laptop, his laptop detects it but cannot read it. What should Ronaldo do to work out what is wrong with the external hard drive? (1)
- 6.3 Ronaldo finds out that he should format his new external hard disk. Why should he do this? (1)
- 6.4 Over the past three years, Ronaldo's laptop has become slower. Mention three things that might be causing this. Also provide a possible solution to each problem. (6)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Describe the difference between primary and secondary storage?		
2.	Discuss online storage?		
3.	List the best storage media for backups and archiving?		
4.	Describe the role and function of various internal computing components, such as the motherboard, CPU, RAM and ROM?		
5.	Interpret advertisements for storage media and computers?		
6.	Do basic troubleshooting on input, output, storage and processing devices?		

SOFTWARE

CHAPTER OVERVIEW

Unit 4.1	The role of application software
Unit 4.2	Collaboration and communication software
Unit 4.3	Compatibility issues
Unit 4.4	Software for physically challenged users
Unit 4.5	Cloud-based applications
Unit 4.6	Online storage

By the end of this chapter, you will be able to:

- Define the role of application software
- Discuss application software in terms of collaboration and communication software
- Define compatibility issues in software
- Describe why software must be updated
- Describe software for physically challenged users
- Explain the advantages and disadvantages of online or cloud storage
- Discuss cloud-based applications with a focus on Google Docs and Microsoft Office 365

INTRODUCTION

In Grade 10, you learned about what software is. Software is the programs that contain the instructions you use to tell the computer what you want it to do. There are two types of software:

1. **Application software** allows you to perform tasks on the computer, for example, playing music or games and creating presentations, spreadsheets and documents.
2. **System software** controls and manages the operations of the computer.

Application software is the most useful type of software and the one that you interact with the most. There are different types of application software, for example:

- Word processors, such as Microsoft Word or LibreOffice Writer.
- Office suites such as Microsoft's Office Suite, which contains programs such as Word, PowerPoint and Excel.
- Internet browsers such as Google Chrome and Mozilla Firefox.
- Movie and media players such as VLC media player and Windows Media Player.

System software manages the computer and the flow of data and information by:

- Translating instructions from the software to the hardware.
- Sending the right instructions to the right hardware to complete the task.
- Receiving information from the hardware and sending it to the software.
- Allowing different applications to communicate with each other.
- Checking the condition of the hardware and the software.

There are also different types of system software, such as operating systems, device drivers and firmware.

In this chapter, you will learn more about application software, specifically collaboration and communication software. You will explore what compatibility issues are and what you can do to work around them. You will also look at what versions, patches and service packs are and what role they play in updating software. We will also highlight application software for physically challenged users.

You will be introduced to online or web-based applications, such as Google Docs and Office 365, and their uses, advantages and disadvantages.

4.1 The role of application software

As you learned, application software (also called an app or application for short) is a computer program or group of computer programs designed for users. It makes using a computer easier. In the past, users would need to enter commands into a text field but now doing anything on your computer is as easy as clicking a button or tapping on an icon.

There are different types of application software. Table 4.1 has a list of some of these types and descriptions of what they do. This list is by no means a full list of application software, since there is software for everything you do on your computer and there are many different variations of the same type of software.

Table 4.1: *Types of application software*

NAME	DESCRIPTION
Word-processor	A word-processing program used to create, edit and manipulate text documents. It can also support images and other graphical elements (such as videos). Example: Microsoft Word
Media player	A media player allows you to watch videos and listen to music that is stored on your computer or streamed from an online source. It can also be used to convert media files to a wide variety of different formats. Example: VLC media player or Windows Media Player
Web browser	An application that allows you to access and browse the World Wide Web. The browser communicates with a web page to display all the web page's features to allow you to navigate around it. Example: Google Chrome, Firefox or Safari
Desktop publishing	Multimedia software that allows you to create, edit and manipulate images and other visual media, such as brochures, logos etc. Example: CorelDRAW, MS Publisher or Adobe InDesign
Email	Communication software that allows you to send and receive electronic communications to and from other people. Example: Microsoft Outlook or Gmail

4.2 Collaboration and communication software

Multimedia and communications applications are some of the most common applications in use today.

MULTIMEDIA SOFTWARE

Multimedia refers to integrating multiple forms of media into a single, often digital, product. This could be a PowerPoint presentation that includes videos and audio clips, a Word document with animations and images, or a web page with interactive elements.

Although images and text are different types of media, multimedia is usually used to refer to things that incorporate video, audio, animated images (such as GIFs) and text.

Multimedia software refers to the programs you use to create and interact with multimedia. There are two broad categories of multimedia software: media players and media creators. Media players are software applications that allow you to play back multimedia files, including audio and video files. Media players can also be used to view still images (such as photographs or illustrations). Some examples of media players are shown in Table 4.2.

Table 4.2: *Examples of media players*

NAME	DESCRIPTION	OPEN SOURCE, FREWARE OR PROPRIETARY
VLC media player	Video and music player that integrates streaming and file conversion.	Open source
YouTube	Video player, only available on the World Wide Web.	Web-based application
iTunes	Video and music player that integrates internet radio broadcasting and mobile device management.	Pre-installed on Apple devices – freeware.
Adobe Flash Player	Used to display interactive web pages and online games as well as audio and video content. Mostly unused now.	Freeware

Multimedia software is also used to create multimedia content, which is any content that uses a combination of text, audio, images, animations, video and interactive elements. There are several different types of multimedia software available, depending on what you would like to create. Table 4.3 shows examples of multimedia software and what they are mainly used to create.

Table 4.3: Examples of multimedia software

NAME	DESCRIPTION	FREWARE EXAMPLES
Adobe Photoshop, InDesign and Illustrator; CorelDRAW	Creating images and graphics, animations and GIFs.	Pixlr, Pablo
Adobe Dreamweaver	Creating websites	WebFlow, Squarespace, WordPress
Adobe Premiere Pro or Adobe Premier Elements	Video editing	OpenShot, Windows Movie Maker, Apple iMovie
Adobe Audition	Sound and music editing	Audacity, Apple GarageBand, Acoustica



Something to know

You may have noticed that in the examples given above, the name Adobe pops up quite often. Adobe Systems was founded by John Warnock and Charles Geschke in 1982, after they left the Xerox company to open a new company to sell their product PostScript.

PostScript was a powerful computer language that described the layout of a page to a printer, allowing the printer to reproduce in hard copy exactly what was on the digital page.

PostScript allowed laser printing to become a reality, since the program could guide the lasers to draw the digital page on the printer's drum correctly. Soon, PostScript became a staple in the advertising industry, since it allowed advertisement creators to quickly print out a copy of their advert to see what it would look like on a page.

Their next revolution was the Type 1 typefaces, which provided digital fonts that could be printed at any resolution. This was ground-breaking, because at the time more elaborate fonts would have to be lettered by hand and could only be printed at higher resolutions or they would lose their detail. The Type library soon became the largest collection of typefaces for personal computers in the world.

The introduction of Adobe Illustrator brought the ability to create detailed and high-quality digital drawings, which changed the face of advertising and marketing forever. Images could be created, manipulated, edited and updated quickly and more easily than traditional by-hand drawings.

Each new version of Adobe's products and each new product they bring out makes creating high-quality multimedia content easier and faster and their products remain some of the most popular multimedia software available.

COMMUNICATION SOFTWARE

Communication software is the group of applications that allow users to share files, text and video or audio messages over an internet connection. It is more than just email; communication software includes **Voice over Internet Protocol (VoIP)** systems and chat programs. Some examples of communication software are:

- Email clients such as Outlook.
- VoIP software such as Skype and RingCentral.
- Chat applications such as Slack and WhatsApp.

Communication software makes it easier for people to communicate and it is often more cost-effective to install and implement than traditional communications methods (such as telephone lines).

Most communication software doubles as collaboration software. This means that it can be used by multiple people to work on the same thing or to share work online. For example, Skype allows you to make video calls and to share your screen with the people on the call with you, allowing them to see what you are working on.



Activity 4.1

1. Multiple choice:
 - a. Which of the following video players is only available on the World Wide Web?
 - A. VLC media player
 - B. YouTube
 - C. Windows Media Player
 - D. iTunes
 - b. Which of the following is not antivirus software?
 - A. NOD32
 - B. Avast
 - C. AVG
 - D. AMG
 - c. What is the file extension for animated images?
 - A. *.jpeg
 - B. *.mp3
 - C. *.gif
 - D. *.mp4

2. Match column B with column A:

COLUMN A	COLUMN B
2.1. An application that allows you to watch videos and listen to music that is stored on your computer or streamed from an online source.	A. Media player
2.2. A program used to create, edit and manipulate text documents.	B. Antivirus
2.3. Communications software that allows you to send and receive electronic communications from other people.	C. Word processor
2.4. A program that provides real-time protection against viruses, malware and other malicious software for your computer.	D. PowerPoint
2.5. Multimedia software that allows you to create, edit and manipulate images and other visual media such as brochures and logos.	E. Paint
2.6. A program that is used for creating and displaying presentations.	F. CorelDRAW
2.7. Basic image manipulation and drawing software that is built into Microsoft Windows.	G. Outlook
2.8. Chat application used for collaboration in business environments.	H. Slack

3. Answer the following questions in your own words:
 - a. Explain what antivirus software is used for.
 - b. List two examples of multimedia software and describe what they are specifically used for.
 - c. Define communications software.
 - d. Why does communications software make it easier for people to communicate?

4.3 Compatibility issues

The huge variety of application software available and all the versions of the same software mean there are bound to be compatibility issues, even when people are using the same kind of software. Compatibility issues come up when users are using the same type of software for a task, such as word processors, that cannot communicate with each other. This could be due to a difference in their versions or because they are made by different companies.

Compatibility issues can be small, for example certain features not working properly in older versions of the same software, but they can also be problematic, such as when a newer version of the software cannot open a document created in an older version.

In Microsoft Word for example, documents created in Word 2016 or 2013 can be opened in Word 2010 or 2007, but some of the newer features (such as collapsed headings or **embedded videos**) will not work in the older versions.

If someone using Word 2016 opens a document created in Word 2010, the document will open in Compatibility Mode. Microsoft Office does this to make sure that documents created in older versions still work properly.



Figure 4.1: Microsoft Word Compatibility Mode

The feature in Figure 4.1 is an example of something called backwards compatibility, which is the ability of newer software to interact with files (or programs or systems) made with older versions of that software. It is usually built into the software and is a way to avoid compatibility issues. Another way to avoid this is to update your software.

UPDATING SOFTWARE

More and more software providers are offering subscription services for their software. An example of this is Microsoft's Office 365, which gives you the full range of Microsoft Office products and access to Microsoft's online productivity tools, such as OneDrive (their cloud storage service). Since Office 365 is a subscription service, you never truly own the software and will need to renew your subscription every year, but the software is updated once a month to make sure that you have access to the latest features.

Making sure that your software is up to date is important as you will be able to get the newest fixes for **bugs** in the software, the most recent virus and malware fixes and the latest features. This is especially important for software such as your antivirus programs and your operating system. Microsoft Windows 10 will let you know when a new version of any of your software becomes available so that you can update it. Some software also has the option to check for and install updates automatically, so you do not need to remember to do so.

TO UPDATE YOUR SOFTWARE

Whenever your computer asks to update a program, allow it to update as soon as possible. If it is not possible to install the update immediately, schedule a time that evening to install the update.

Windows Update



You're up to date

Last checked: Today, 08:28

Check for updates

Feature update to Windows 10, version 1903

The next version of Windows is available with new features and security improvements. When you're ready for the update, select "Download and install now".

[Download and install now](#)

Figure 4.2: Microsoft Windows automatically installs updates

If the application informs you that an update is available with a link, download the newest version and install the downloaded program. This will update the program to the newest version without changing your settings.

If a program you use often does not inform you about an update, search the internet once or twice a year to see if a new version has been released. If it has, download this version and install it to update your software.

PATCHES AND SERVICE PACKS

A patch is a program that makes changes to software installed on a computer. Software companies issue patches to fix bugs or security problems in their programs or add new functions to the software. Microsoft refers to their patches as hotfixes.

A service pack (SP) is a collection of updates and fixes for an operating system or a software program. SPs are usually made up of patches. Many of these patches are often released before a larger SP, but the SP allows for an easy, single installation.

An installed SP also usually updates the version number for Windows. This is the actual version number, not the common name, such as Windows 10.

SPs often include new features as well as fixes. This is why one version of a program or operating system can be very different to another on a different computer.

Most, if not all, operating systems and software programs provide SPs free of charge as either a manual update from the developer's website, or through an auto-update feature within the program or operating system.



Something to know

Since releasing Windows 10, Microsoft has moved away from SPs and instead releases regular major features updates.



Activity 4.2

1. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. Microsoft Office 2016 is an example of a subscription service.
 - b. Microsoft refers to their patches as hotfixes.
 - c. An installed service pack (SP) also usually updates the version number for Microsoft Office.
 - d. Most SPs are provided for a fee.
2. Answer the following questions in your own words:
 - a. Explain what a compatibility issue is.
 - b. Describe what backwards compatibility is.
 - c. Why are more software companies offering subscription services for their products?
 - d. Elaborate on the main difference between buying software and using a subscription service.

4.4 Software for physically challenged users

In this unit, we will take a closer look at the software incorporated into most software for able-bodied users that makes it easier for physically challenged users to use.

Each assistive technology input and output device comes with specialised software to interpret the input correctly to give the correct output. For example, text-to-speech software converts text on a page to speech, allowing visually impaired users to “read” the page. Some of the more complex input and output devices, such as gesture-controllers and sip-and-puff devices, need specific software to work correctly. This software is usually included with the devices.

More recently, the developers of operating systems for smartphones and computers have begun to include accessibility software in their operating systems. Microsoft’s Windows 8 and 10, Apple’s OSX and Google’s Android operating systems all have settings and built-in applications designed for physically challenged users.

While alternative input and output devices and accessibility software are important components in giving physically challenged users the opportunity and ability to use computers, these devices cannot work properly if the information they are interacting with has not been put together with accessibility in mind.

This is especially important when it comes to accessing the internet. Screen readers can struggle with interpreting websites that have not been designed for visually impaired readers. Since websites and web pages are mostly designed as a visual experience, users who cannot see these visual elements struggle to interact with the sites.

There are, however, ways that designers and developers can make their websites more accessible to visually impaired or blind users, for example:

- Giving links and buttons descriptive labels (saying “click here to sign up” instead of just “click here”).
- Giving images and graphics good descriptions and *Alt Text*.
- Using headings to organise the content on a website’s pages.
- Using descriptive titles for pages.
- Closed captions, which describe all the audio elements of a video and not just the dialogue, can also help deaf and hearing-impaired users watch videos and learn from them.

Almost all modern operating systems contain a series of special settings and functions to make using a computer easier for physically challenged users. These are usually referred to as accessibility settings and can include things such as:

- Colour settings for colour blind people to make browsing the web easier.
- Magnifying the screen to make it easier to read for partially blind users.
- Text-to-speech software to read the typed text out loud to blind users.
- Displaying visual notifications instead of having a notification sound to alert users who are hard of hearing to potential problems or updates for their computer.
- Speech recognition software to allow users to control their computers using their voices.



USING A SCREEN READER



<https://youtu.be/7Rs3Ypsnfol>

You can see what range of accessibility options your computer has by clicking on the Windows icon, searching for *Control Panel* and clicking on *Ease of Access*.

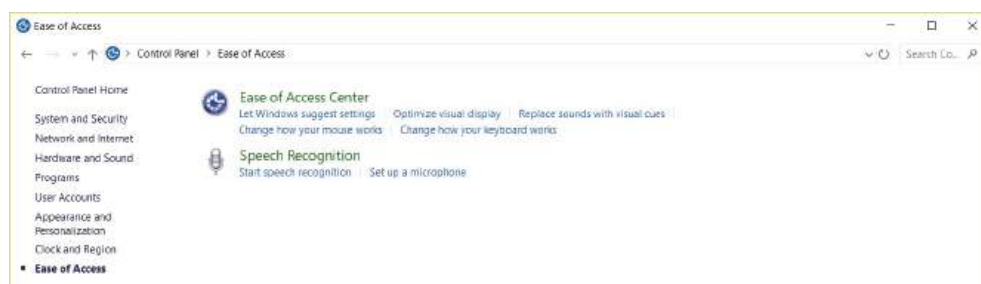


Figure 4.3: *Ease of Access menu in Windows 10*



Activity 4.3

1. Match Column B with Column A:

COLUMN A	COLUMN B
1.1. Special settings and functions to make using a computer easier for physically challenged users.	A. Screen reader
1.2. Device used to interpret websites for visually impaired readers.	B. Monitor
1.3. Click on <i>Ease of Access</i> on the <i>Control Panel</i> to view...	C. Accessibility software
	D. Accessibility options
	E. Accessibility settings

2. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. To find accessibility options, you need to click on Ease of Use.
 - b. Speech recognition software reads text aloud for blind users.
 - c. Closed captions can help deaf and hearing-impaired users watch videos.
3. Answer the following questions in your own words:
 - a. What are accessibility settings? Give two examples.
 - b. How do you access the accessibility features in Windows 10?
 - c. Give one example of software for physically challenged users.
 - d. Why is it important to design websites with accessibility in mind?
 - e. List four ways in which website developers can make their websites more accessible.

4.5 Cloud-based applications

With the rise in internet access and smartphone use, more people are turning to online or cloud-based applications to get their work done, since they can access these programs wherever they are as long as they have an internet connection. Cloud-based applications are any applications that are accessed using an internet connection instead of being stored in a device's memory.

Gmail, Google's online email client, was more popular in 2018 than Microsoft's Outlook. Online service providers are also offering cloud-based applications to businesses.

In this unit we will look at two web-based applications, namely Google's G Suite and Microsoft Office 365. We will discuss the advantages and disadvantages of each and how you can use them.

GOOGLE G SUITE

Google's G Suite is a free, online-only office suite. You can use it (with Google Docs and its companion applications Google Sheets and Google Slides) to create more or less the same documents you can create using more traditional desktop office suites (such as Microsoft Office). Google G Suite is offered with Google Drive, Google's cloud storage platform, and Gmail.

Google Docs has several advantages, for example:

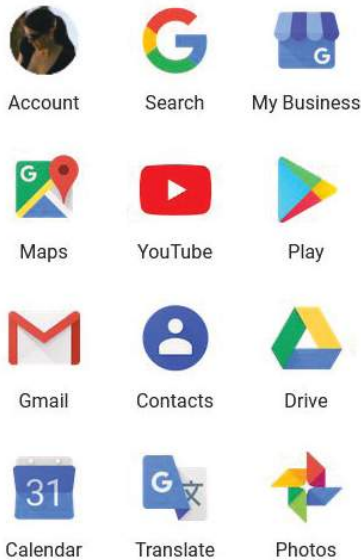
- It is easier to share and work with others in real time and multiple people can work on the same document at the same time.
- While Google Docs does have an offline editing option, it will only save your changes once you are connected to the internet again.
- The software is always up to date and does not require you to download or install anything.
- You can access your documents anywhere you have internet access and access to your Google account.
- Your documents do not take up space on your computer's storage. Instead, they are saved on the cloud.
- Because the documents are saved online, your documents are backed up automatically.

However, Google Docs also has its disadvantages, such as:

- It does not have the full range of word-processing features, e.g. you are limited to a set number of styles and formatting. This means that you cannot customise your documents as much as you can with other cloud-based applications.
- As with anything stored on the internet, your documents are vulnerable to attack and you do not have full control over the security of your documents.

ACCESSING GOOGLE DOCS

To access Google Docs, you will need to go to Google Drive. You can do this by clicking on this icon  in your Gmail account, followed by clicking on *Google Drive*. You need to set up a Gmail account to get access to Google Docs.

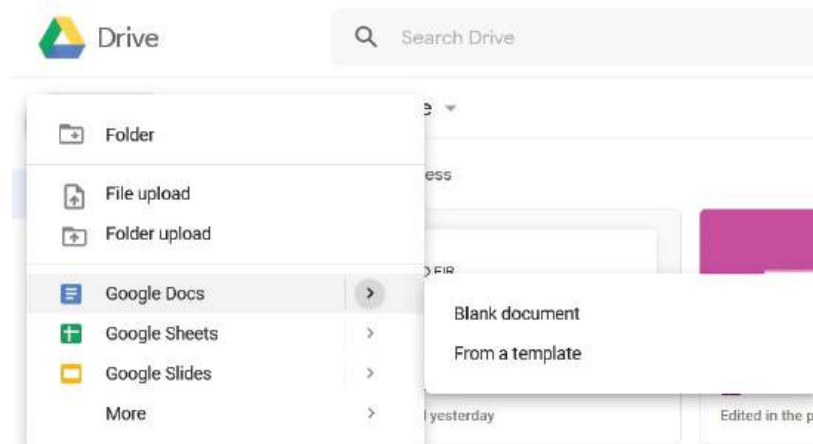


Alternatively, you can visit <https://drive.google.com>.

Once you are in your Google Drive account, click on *New*.



Click on *Google Docs* and choose *Blank document* or *From a template* to get started.



MICROSOFT OFFICE 365

Microsoft Office 365 is a subscription-based software service that combines traditional, locally stored software (the Office Suite) with cloud-based applications and storage. As part of your Office 365 subscription, you get access to Microsoft's OneDrive and online versions of Microsoft's word-processing, presentation, spreadsheet and database-management software.

You are also able to share this software across multiple devices you own, for example a laptop, smartphone and tablet.

The advantages and disadvantages of Microsoft Office 365 are very similar to the advantages and disadvantages of Google Docs. While Office 365 does come with more features, you still have to pay to use them and you will need to renew your subscription every year.

Table 4.4 shows how Google G Suite compares to Microsoft Office 365.

Table 4.4: *Comparison between G Suite and Microsoft 365*

GOOGLE G SUITE	MICROSOFT OFFICE 365
Free, online-only office suite	Subscription-based software service
You can use it (with Google Docs and its applications Google sheets and Google slides) to create more or less the same documents you can create using more traditional desktop office suites (such a Microsoft Office).	Combines traditional, locally stored software (the Office Suite) with cloud-based applications and storage.
Software is always up to date and does not require you to download or install anything.	While Office 365 does come with more features, you still have to pay to use them and you will need to renew your subscription every year.
Google's G Suite includes Gmail, Google Drive, Google Docs, Sheets, Slides, Calendar, Keep, Hangouts, and other Google apps.	Microsoft's Office 365 includes Outlook, OneDrive, Word, Excel, PowerPoint, One Note, Skype, and other Microsoft apps.



Activity 4.4

- Multiple choice:
 - Which of the following is Google's online storage service?
 - Google Drive
 - Google Docs
 - Chrome
 - OneDrive
 - Which of the following is NOT part of the Microsoft Office 365 package?
 - Outlook
 - OneNote
 - PowerPoint
 - Notepad
 - Which application does not need an internet connection to work?
 - Notepad
 - Google Docs
 - Office 365
 - Google Maps
- Match Column B with Column A:

COLUMN A	COLUMN B
2.1. Which online office suite has the advantage of being used with any operating system and web browser?	A. OneDrive
2.2. Which company's online services have different payment packages?	B. Google Drive
2.3. Microsoft's version of a file hosting service that synchronises with Microsoft's Office Online services.	C. OneNote
	D. Microsoft Office 365
	E. Google Docs

... continued



Activity 4.4

... continued

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. You have access to a full range of formatting options in Google Docs.
 - b. Microsoft Office 365 is a free service.
 - c. You can use Office 365 on multiple devices.
 - d. Google Docs is a free, online-only office suite but you pay for extra storage.
 - e. Google Drive is separate from your Gmail account.
4. Answer the following questions in your own words:
 - a. Explain what a cloud-based application is.
 - b. List the steps taken to access Google Drive and open a spreadsheet.
 - c. List two advantages Google Docs has over Microsoft Word.

4.6 Online storage

Online storage is becoming more popular as access to the internet grows. Online storage is also called **tertiary storage** or cloud storage and uses websites and online portals to store information on the World Wide Web. Examples of online storage include services such as Google Drive, Dropbox and Apple iCloud.

While storing information online seems like you are sending your information into the cloud, you must remember that this information does get stored on a physical computer somewhere, usually on a server in a server farm.

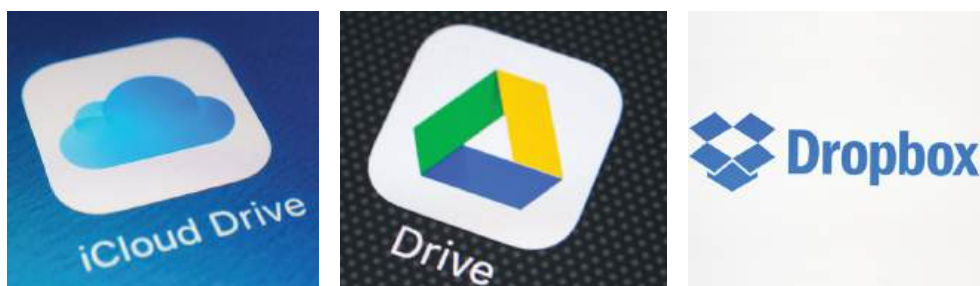


Figure 4.4: Online storage services include Apple iCloud, Google Drive and Dropbox

Online storage has several advantages and disadvantages.

Some of the advantages of online storage are:

- Allows you to access your information anywhere.
- Frees up physical storage space on storage devices.
- A reliable way to back up important or critical information.
- Data stored online can be shared more easily

A number of disadvantages of online storage:

- It is vulnerable to attacks (hacking) and distribution.
- May have **bandwidth** or storage limits.



Activity 4.5

1. Multiple choice:
 - a. Which of the following is not an example of online storage?
 - A. iCloud
 - B. Google Drive
 - C. Google Docs
 - D. Dropbox
 - b. Which of the following has potentially the largest storage space?
 - A. Primary storage
 - B. Cloud-based storage
 - C. External HDD
 - D. SSD

... continued



Activity 4.5

... continued

2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1. Information gets stored on a physical computer, usually in a server farm.	A. Internet access
2.2. Cloud-based storage is also considered as ...	B. PC
2.3. The most important thing needed to access online storage; this is also a disadvantage.	C. Tertiary storage
	D. HTTP
	E. Online storage

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
- Online storage has limited free space and you pay to get more space.
 - Dropbox is only compatible with the Microsoft Windows operating system.
 - Spotify songs are stored on a cloud-based storage system.
4. Answer the following:
- Explain how your data can potentially be lost when using cloud storage.
 - List THREE advantages of cloud-based storage.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 What is the definition of software? (1)
- Programs that allow you to perform specific tasks on the computer.
 - Programs that control and manage the operations of the computer.
 - Programs that contain the instructions you use to tell the computer what you want it to do.
 - Programs that allow different applications to communicate with each other.
- 1.2 Which of the following is NOT an example of application software? (1)
- Microsoft Excel
 - Norton Antivirus
 - Windows XP
 - Mozilla Firefox
- 1.3 Which of the following is multimedia software? (1)
- Microsoft Paint
 - Microsoft Access
 - Microsoft Edge
 - Notepad
- 1.4 Which of the following is an example of system software? (1)
- WordPad
 - Windows Defender
 - Google Chrome
 - Windows Media Player
- 1.5 Hotfixes are also known as _____. (1)
- Traces
 - Hotspots
 - Patches
 - Service packs

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- You can own subscription software as long as you pay for it. (1)
- Subscription software can update itself automatically. (1)

... continued

REVISION ACTIVITY

... continued

- c. Service packs are updates for system software. (1)
- d. Google Docs has all the features that Microsoft Word has. (1)
- e. Multimedia software refers to the plans you use to create and make multimedia. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1.J). (5)

COLUMN A	COLUMN B
1. Software used to create 3D models.	A. Adobe Photoshop
2. Software used to play movies and music.	B. Microsoft Excel
3. Software to email people online.	C. Google Drive
4. Software used to edit photos and GIFs.	D. Google Chrome
5. Software for data storage.	E. VLC media player
	F. NVIDIA GeForce Experience
	G. Windows Defender
	H. Dropbox
	I. Microsoft Outlook

QUESTION 4: MEDIUM AND LONG QUESTIONS

- 4.1 Describe what backwards compatibility is. (2)
- 4.2 Explain briefly what accessibility settings are. (2)
- 4.3 Give two examples of accessibility settings. Mention which physical challenge could benefit from this setting. (4)
- 4.4 Identify the compatibility issue for the following scenarios. (4)

SCENARIO	COMPATIBILITY ISSUE
1. While playing a newly installed game, you realise that at certain points on the game map the controllers are not responding the way they should.	
2. The new functions advertised by an application are not working or showing on my device.	
3. As soon as I open a Word document on the school computer, a window pops up saying that certain characters will not be displayed.	
4. Every time I use a calculator app on my laptop and press the multiply button, it adds the values instead.	

- 4.5 Your teacher has asked you to create a presentation that includes at least one video, some audio clips and an animation. Which three software programs will you use to make these multimedia items and why? (6)
- 4.6 Describe the difference between a service pack and a patch. (4)
- 4.7 Provide two reasons why it is important to update your software regularly. (2)
- 4.8 Give one advantage Microsoft Word has over Google Docs. (1)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Define the role of application software?		
2.	Discuss application software in terms of collaboration and communication software?		
3.	Define compatibility issues in software?		
4.	Describe why software must be updated?		
5.	Describe software for physically challenged users?		
6.	Explain the advantages and disadvantages of online or cloud storage?		
7.	Discuss cloud-based applications with a focus on Google Docs and Microsoft Office 365?		

HARDWARE, SOFTWARE AND COMPUTER MANAGEMENT

CHAPTER 5

CHAPTER OVERVIEW

Unit 5.1 Basic system requirements

Unit 5.2 Software installation

Unit 5.3 File management

By the end of this chapter, you will be able to:

- List and describe basic system requirements and how they relate to software
- Discuss the difference between installing software from an online download and a portable storage medium
- Describe the installation process
- Explain how to manage files and folders on a computer

INTRODUCTION

In this chapter, you will learn what basic system requirements are and how they link to software. You will learn how to install software from a portable storage medium (such as a CD) and from an internet download, as well as dig deeper into file management and what file attributes are. You will also learn how to **import** and **export** files and how to search for files.

5.1 Basic system requirements

System requirements are the minimum necessary specifications that you will need to make sure that the software runs smoothly and does not overwork the hardware on your computer. It is basically a list of what you need to make sure that a game or program runs properly.

If your computer does not meet these requirements, you may not be able to install the software or, if you can install the software, it may run slowly. The system requirements for software are usually printed on the package that the software comes in or are listed on the software's website.

Figure 5.1 below shows the list of system requirements for installing Photoshop.

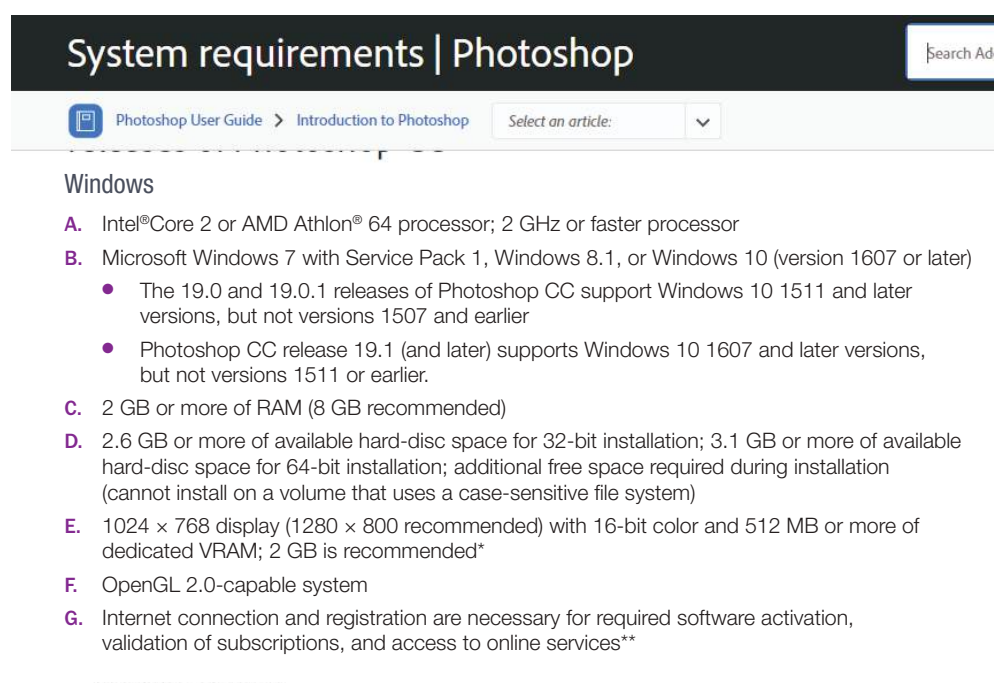


Figure 5.1: Adobe Photoshop system requirements

As you can see, the software makes several recommendations, including:

- A. The type of CPU you will need (Intel® Core 2 or AMD Athlon® 64 processor operating at 2 GHz or faster).
- B. The operating system you should be using (Microsoft Windows 7, 8.1 or 10).
- C. How much RAM your system should have (a minimum of 2 GB but 8 GB is recommended).
- D. How much space should be on your hard drive.
- E. The size of your display (or screen).
- F. Whether or not you will need an internet connection.

Knowing what the system requirements are for the software you will be using most often will help you decide what computer to buy when you are shopping for one. Most of the information you will need will be on a label on the computer or will be listed in an advertisement for the computer.

System requirements will also let you know if you need to purchase any additional hardware (such as microphones when using audio recording software). Hardware items, such as

printers, also come with system requirements. For example, a printer may only be compatible with Windows 8.1 and 10 or Apple's macOS and cannot be connected to a computer running Windows 7.

For your computer to be able to run all the newest gaming software, you would need to look at the graphics card. A GPU is a programmable logic chip to render images, animations and video for the computer's screen. GPUs can be located in plug-in cards, in a chipset on the motherboard or in the same chip as the CPU. The gaming software's system requirements will recommend the size of the additional GPU.

If your computer does not meet all the requirements to run specific software or communicate with a specific device, you will not necessarily have to buy a whole new computer. It is possible to upgrade your RAM, GPU or CPU. You can, for example, upgrade your operating system or you can install the latest device drivers to make sure that your computer and hardware can communicate with each other.



Activity 5.1

1. Answer the following questions pertaining to the advertisements below:

Features for **DELL 39 cm (15.6") Inspiron 3552 Intel Celeron Laptop**

Operating System:	Windows 10 Home
Processor:	Intel Celeron - N3060 (Up to 2.48 GHz)
Memory:	4 GB RAM
Storage:	500 GB hard disk drive
Graphics:	Intel HD
Optical Drive:	Yes
Display Size:	39 cm (15.6")
Display Resolution:	1366 x 768
Touch Display:	No
HDMI Input:	Yes
USB Ports:	Yes
SD Card Reader:	Yes
Connectivity:	Wi-Fi / Ethernet / Bluetooth

Photoshop system requirements

Search Adobe Support

[Adobe Photoshop User Guide](#) > [Introduction to Photoshop](#) > [Select an article](#)

Minimum system requirements for Photoshop

Windows

	Minimum requirement
Processor	Intel® or AMD processor with 64-bit support*, 2 GHz or faster processor
Operating system	Microsoft Windows 7 with Service Pack 1 (64-bit)*, Windows 10™ (Version 1709 or later)
RAM	2 GB or more of RAM (8 GB recommended)
Hard disk space	33 GB or more of available hard-disk space for 64-bit installation; additional free space required during installation (cannot install on a volume that uses a case-sensitive file system)
Monitor resolution	1024 x 768 display (1280x800 recommended) with 16-bit color and 512 MB or more of dedicated VRAM; 2 GB is recommended***
Graphics processor acceleration requirements	OpenGL 3.0-capable system
Internet	Internet connection and registration are necessary for required software activation, validation of subscriptions, and access to online services

ON THIS PAGE

- [Minimum system requirements for Photoshop](#)
- [Language versions available for Photoshop](#)
- [Graphic processor requirements](#)
- [System requirements for Adobe Camera Raw](#)

Applies to Adobe Photoshop
Last Published: April 3, 2019

Win Web page help icon

- a. Does the DELL laptop have the minimum requirements for Windows 10? How can you be sure without looking at the minimum hardware needed for Windows 10?
- b. What does OS mean in the *Call of Duty: Modern Warfare 2* game requirements?
- c. What does *display resolution 1366 × 768* mean?

... continued



Activity 5.1

... continued

- d. Describe what is meant by *Touch Display*.
 - e. Explain what 32-bit means.
 - f. Decide if the *Call of Duty: Modern Warfare 2* and Photoshop in Figure 5.1 can be installed on the laptop and add relevant comments.
 - g. What does VRAM stand for?
 - h. State which component of the computer will be the easiest and cheapest to upgrade.
 - i. List the different processors' names in the adverts and sort them from fastest to slowest.
2. Write True or False next to the question number. Correct the statement if it is FALSE. (You may not simply use the word NOT to change the statement.)
- a. If your hardware does not meet the system requirements, it means that the software will not work.
 - b. Certain programs can run even if you have no RAM.
 - c. Certain programs cannot work if you do not have the right operating system.
 - d. The motherboard is the core component in a computer that brings all the different aspects of the computer together.

5.2 Software installation

Once you have worked out what requirements you need to install software, you will need to begin the installation process. Before installing the software, you need to:

1. Download the software files from a website; or
2. Purchase a CD or DVD from a store.

After downloading the installation software, you can install it. Usually this installation software is an **installation wizard** to assist you with the installation process.

ONLINE DOWNLOAD

The most common way to get new software is to download it from the internet. To do this, you must go to the website where the software is available and download the files. Make sure that you download software from trusted publishers or verified retailers only. This ensures that you are getting the software you want and that you do not accidentally download malware or a virus.

You will usually use a credit card to buy the software to download. Once you have purchased the software, you will click on the download link to begin the process.

You can also buy software from app stores, such as Google's Play Store, Apple's App Store or the Windows Store. The software you are looking for is usually sorted according to its category, for example, gaming, word processing or design.

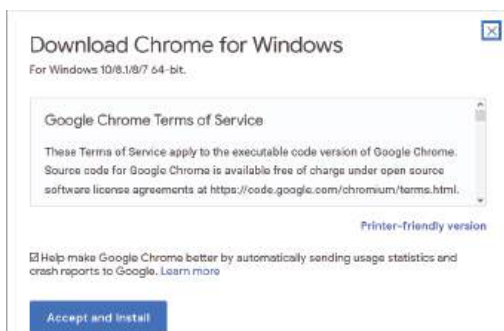
DOWNLOADING CHROME

To download software from the internet, do the following:

1. Go to <https://www.google.com/chrome/> (or the site where the software is available to download).



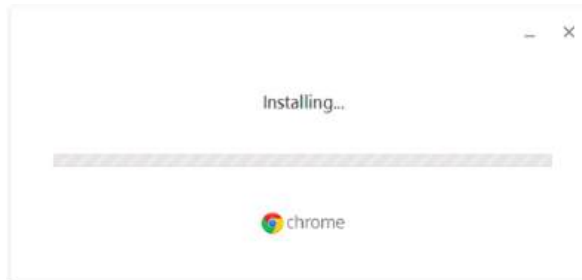
2. Click on the *Download* button. A dialog box will appear. Accept the *Terms of Service* and make sure that you are downloading the correct version for your computer.



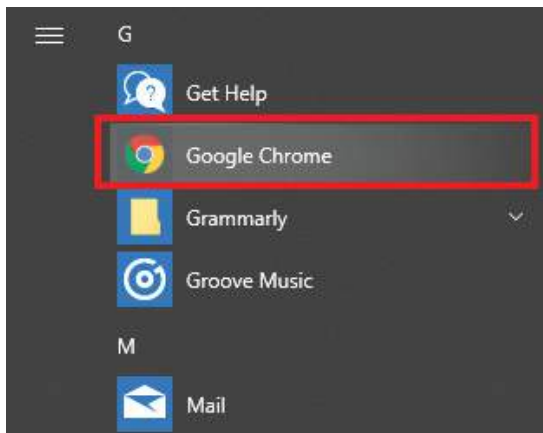
... continued

DOWNLOADING CHROME

3. Follow the instructions to install the software.



4. Open the application from the *Start* menu.



PORTABLE STORAGE MEDIUM INSTALLATION

Some software can be installed using the files found on a CD or DVD. This includes software such as the operating system, Microsoft Office (not Microsoft Office 365) and some games.



Figure 5.2: An example of software on a portable storage medium

To get this software, you will need to go to a store and buy the product and bring it home with you to install. You do this by inserting the disk into your computer's optical disk drive (its CD or DVD player). The installation program will usually start automatically but you can also look for the setup program to start the installation.

HOW TO INSTALL SOFTWARE

Once you have downloaded the software or inserted the disk, the installation wizard will guide you through what to do. While the installation wizard does not handle all the aspects of the installation, since these may be different from one program to the next, the process is very similar for all software.

SOFTWARE STEPS FOR INSTALLATION

You will usually start by either clicking on the downloaded file or inserting the disk and then letting the program setup file run (usually called *Setup.exe* or *Install.exe*).

STEP	DESCRIPTION
Accept the license agreement	Most software cannot complete the installation if you do not agree to the manufacturer's terms of use. This license agreement is a legally binding contract between you and the manufacturer and outlines what you can and cannot do with the software.
Enter the product key/activation code/registration number	The product key is the way that the manufacturer links the software to you, stops the software from being installed on more devices than the licensing terms agree to and prevents piracy. It also links you to any online support available.
Choose the installation location	The installer will suggest a default folder. It is usually best to stick to this suggestion. For example, if your installer suggests installing all your software on the C:/ drive, you will know where to look for a program.
Choose the type of installation you want	You can usually choose between the following types of installation: • Typical installation, which installs the components that are used most often. • Custom installation, where you can choose which components to install. • Full installation, which installs all the components of the software. This is not always an option in an installation.
Install extras	Some software comes with extra software bundled in (such as McAfee antivirus with some Microsoft products). In this step you can choose which extra software to install. Often these extras are ticked by default and you will have to untick them to not install them. You should always check which extras are installing to your computer and untick the ones you do not want.
Check for updates	Once a program has installed, the installer may ask if you want to check for the latest version. This is usually done with software that is installed from a disk. Downloaded software is often more up to date.
Register your product	Some programs require that you register on their online portal so that the company can contact you with news and notifications of updates. You should be allowed to skip this step if you do not want to be contacted by the manufacturer.
Add shortcuts	The last step in the installation process is to add a shortcut. Most software will ask you if you want a shortcut to the program on your desktop. If you do not, deselect this option.
Take a tour	Some software offers a tutorial the first time you open the program. These tutorials will usually highlight important features and how to use them. If it is the first time you are using a specific program, you should work through the tutorial before getting started.

HOW TO UNINSTALL SOFTWARE

Uninstalling software is when you remove the program and all its files from your computer.

UNINSTALL A PROGRAM

To uninstall a program in Windows 10:

Step 1: Click on the *Windows* icon.



Step 2: Click on *Settings*.



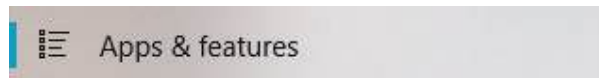
Step 3: Click on *Apps*.



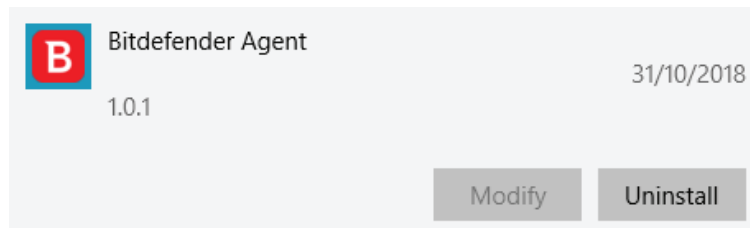
Apps

Uninstall, defaults, optional features

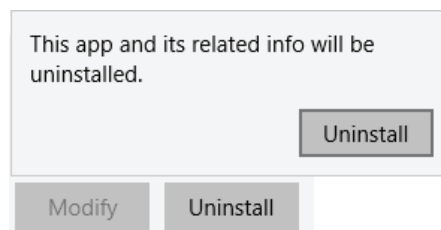
Step 4: Click on *Apps & Features* in the left pane. A list of all installed apps will appear.



Step 5: Click on the one you want to uninstall and then click on the *Uninstall* button (if this button is greyed out, it is a system app and cannot be uninstalled).



Step 6: Click on *Uninstall* in the pop-up window that appears and wait for the system to uninstall the app.



Activity 5.2

1. Write True or False next to the question number. Correct the statement if it is FALSE. (You may not simply use the word NOT to change the statement.)
 - a. You completely eradicate the associated files and folders when you uninstall a program.
 - b. Some installation software files are sold on a flash disk.
 - c. The installation wizard will guide you through what to do when installing software.
 - d. Most hardware is already installed on the Microsoft operating system, thus most hardware are plug-and-play devices.
 - e. Most primary hard drives are saved with the drive location of E:\.

5.3 File management

When you are setting up a file management system, there are several key things you need to look out for. You will need to know what types of files you are creating and how big those files are (so that you know how much of your computer's resources they are using).

It also helps you to understand certain attributes that files have, such as hidden and read-only, and how to import/export files and how to search folders and drives for a specific file. This section will look at these aspects of files.

FILE TYPES

In Grade 10, you learned about file extensions and how they give you an idea of what content is likely to be in the file. File extensions usually link to a specific file type. For example, seeing a file that ends in .docx will let you know that it is a Microsoft Word document. File types indicate the specific characteristics of a document and will let the operating system know which program to use to open that file.

If you were to try and open an Adobe Photoshop document using Microsoft Excel, it would not work. Some file types are associated with specific programs, while others are standard file types that can be opened by a range of programs that support that file type. File types are not file formats, however. The file format usually describes the structure of the file and what it contains. Table 5.1 shows a list of file types and if they are program specific or standard.

Table 5.1: *File types*

FILE EXTENSION	NAME	TYPE
*.pdf	Portable Document Format	Adobe Acrobat Reader specific
*.docx	Microsoft Word document	Microsoft Word specific
*.rtf	Rich text format	Standard or open format
*.txt	Text file	Standard or open format
*.xml	Extensible Markup Language file	Extensible Markup Language
*.mp3	MP3 audio file	Standard or open format
*.flv	Animate video file	Standard or open format
*.gif	Graphic interchange format	Standard or open format
*.jpg	Image file	Standard or open format
*.accdb	Access database file	Microsoft Access specific
*.xlsx	Excel spreadsheet file	Microsoft Excel specific

When you are organising files, you can either organise them by file type or by subject.



Something to know

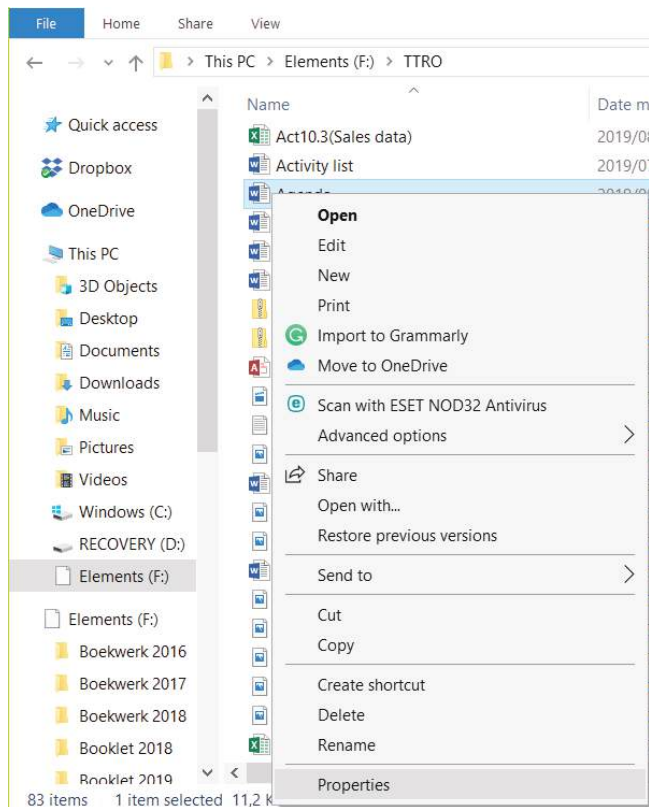
You can find a complete list of all file types at FileInfo.com (<https://fileinfo.com/>).

PROPERTIES

The file *Properties* window shows you a range of information about a file, such as its file type, size and when it was last modified. It can also show you information on where a photograph was taken (its [geo-location](#)) and even what type of camera was used to take it. The *Properties* window will also give you information on how you can manipulate the file.

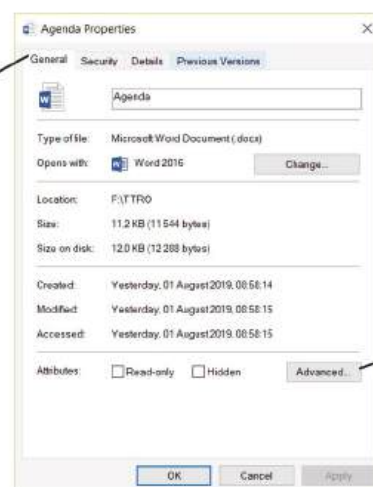
FILE PROPERTIES

1. To see the properties of a file, right click on the file name and select *Properties*.



2. Click on *Properties*.

The General tab shows you the basic information about the file, including file name, size and location on the F:\drive.



Clicking on Advanced will show you more attributes for the document.

The *Properties* dialog box will give you information about the file, as you can see in the example above.

The *General* tab shows you the basic information about the file, including its name, size and location on the C:\ drive. It also shows attributes such as “read-only” and “hidden”. Clicking on *Advanced* will show you more attributes for the document. We will discuss these in more detail later on in this unit.

The *Security* tab allows you to change what access the system and other users have to the file. The *Details* tab gives you more in-depth details about the file, including its word count and what template was used to create it. The *Previous Versions* tab allows you to see if there are other versions of the file.

FILE ATTRIBUTES

File attributes are settings linked to computer files that give or deny certain rights to how a user or the operating system can access that file. There are several types of file attributes, but the most important are “read-only” and “hidden”.

Read-only files cannot be changed, edited or saved. They can only be opened and read. This is done to protect the data in the file from being edited accidentally. Hidden files are any files with the “hidden” attribute turned on. This means that these files are invisible while browsing through folders – you cannot see them without specifically allowing all of them to be seen.

IMPORT AND EXPORT

Most computer programs have their own way to organise and encode data into the files they save. Therefore, they can recognise their own files, but not necessarily those from another program. To be able to send or receive such files, it has to export or import the files. To export a file means to convert it into a format other than the one it is currently in. To import a file is to bring it from another program into the one you are using. This function allows different computer programs to read each other's files.

Applications such as Access or Excel allow you to import or export your data from one file type to another. A good example of this is importing your Outlook data (.ost files) into Excel (.xlsx) or Word (.docx) files. This allows you to make a backup of your data.

You can also use this function to convert files from one file type to another. Importing files also allows you to quickly and easily move your files from one computer to another, especially if you are moving from an Apple product to a computer that uses the Microsoft Windows operating system.

SEARCH

You can use *File Explorer* to search for a file you saved. This is especially helpful if you have many files or you have forgotten where a particular file is stored. If you remember the file name, you can simply enter that in the search box in *File Explorer*.

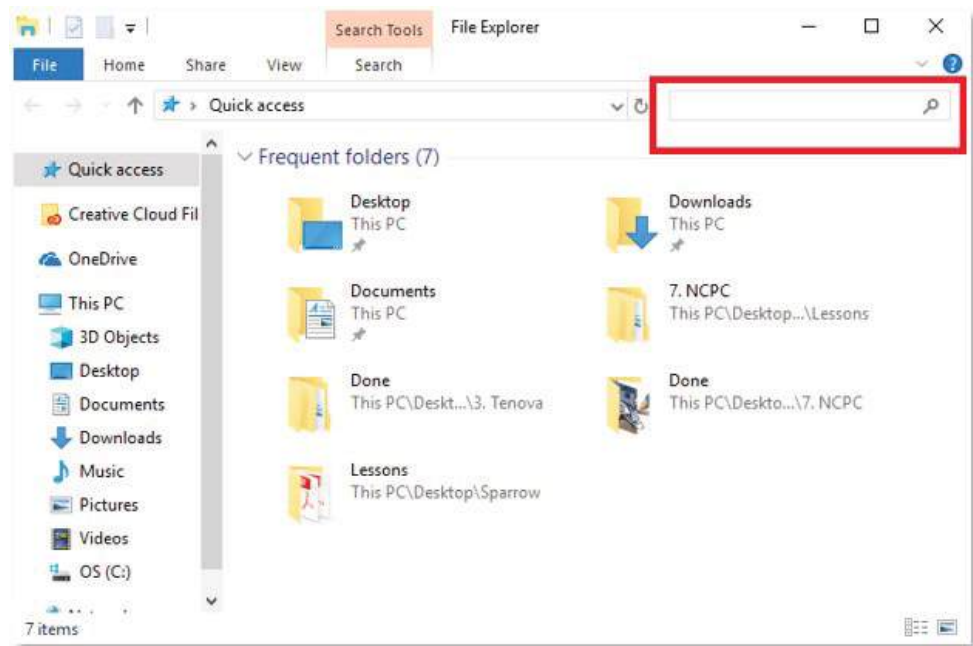


Figure 5.3: The search box in File Explorer

You can also search by file type or size. Using an asterisk wildcard (*) will let you search for groups of files, for example:

- *.txt searches for all text files.
- Home* will search for all files that start with “home”, such as homework.docx or homedesign.psd.



Activity 5.3

1. Multiple choice:
 - a. Which attribute allows the user to view but not edit the file?
 - A. Hidden
 - B. Security
 - C. Read-only
 - D. Sharing
 - b. Which one of the following file extensions is not used in typing a report?
 - A. .txt
 - B. .docx
 - C. .pdf
 - D. .png
 - c. Which of the following file extensions is not part of the Microsoft Office 365 package?
 - A. .txt
 - B. .docx
 - C. .xlsx
 - D. .pttx

... continued



Activity 5.3

... continued

2. Match Column B with Column A:

COLUMN A	COLUMN B	ANSWER
2.1. File extension for an animation.	A. .gif	
2.2. File extension for a picture.	B. Import	
2.3. Term used to describe information on where and when a photograph was taken.	C. .mp3	
2.4. To convert a file into the format required by the application being used. For example, most word-processing programs can ... documents created in other word processors.	D. .jpg	
2.5. To save a copy of the current open document, database, image or video into a file format required by a different application.	E. Geo-satellite	
	F. Geo-location	
	G. Export	

3. Write True or False next to the question number. Correct the statement if it is FALSE. (You may not simply use the word NOT to change the statement.)

- a. You can find a complete list of all file types at FileInfo.com.
- b. The file extension can indicate what type of data is on the file as well as what program is needed to open the file.
- c. The most important file attributes are security and sharing.

4. Answer the following in your own words.

- a. How do file extensions help you identify the type of document?
- b. Give two examples of standard file types.
- c. What does the *Properties* window tell you about a file?
- d. Explain how you can see the file properties of a document.
- e. What are read-only and hidden files?
- f. Explain how you search for a file in Windows 10.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is not a system requirement? (1)
- A. Type of CPU
 - B. Type of RAM
 - C. Type of operating system
 - D. Storage capacity
- 1.2 A hidden file is _____. (1)
- A. A file that cannot be changed
 - B. A file that cannot be edited
 - C. A file that will not display
 - D. A file that cannot be saved

... continued

REVISION ACTIVITY

... continued

- 1.3** A software installer will use _____. (1)
A. A password
B. An online verification system
C. A CD
D. An installer wizard
- 1.4** Which of the following is an example of a program setup file? (1)
A. Word.docx
B. Photoshop.psd
C. Chrome.exe
D. Notepad.rtf
- 1.5** A license agreement indicates the manufacturer's _____. (1)
A. Terms of use
B. Contact details
C. Rules and regulations
D. Terms of agreement

QUESTION 2: TRUE OR FALSE

- 2** Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
- a.** Before a person can use installed software, he or she must accept the license agreement. (1)
b. Uninstalling software involves removing the program and all its files from your computer. (1)
c. File properties are settings linked to computer files that give or deny users access to certain rights. (1)
d. Importing a file converts it from one format to another. (1)
e. Windows Explorer is used to search for files on a computer. (1)

QUESTION 3: MATCHING ITEMS

Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. Where you should go to find information about a file.	A. System requirements
2. A file that can be opened and edited by Adobe Acrobat.	B. Details tab
3. The properties a gamer must look at to see if a game will run on their computer.	C. Read-only file
4. Where you should go to find out what access the system and other users have to the file.	D. Adobe Illustrator
5. A file that can only be read and not edited or saved.	E. .pdf
	F. Export
	G. Security tab
	H. Check for updates

QUESTION 4: MEDIUM QUESTIONS

- 4.1** Name two methods that can be used to install software. (2)
4.2 What is the advantage of using a custom install method? (1)
4.3 What is the difference between exporting a file and importing a file? (2)

... continued

REVISION ACTIVITY

... continued

4.4 Give two examples of file extensions for the following file types:

- a.** Audio files (2)
- b.** Text files (2)
- c.** Video files (2)
- d.** Picture files (2)
- e.** Document files (2)

QUESTION 5: SCENARIO-BASED QUESTIONS

The Civvie Computer Club meets once a week to discuss new and trendy computer-related topics. The club members frequently need to install new versions of software by using a CD. Installing software from a CD is easy since you just need to follow the installation wizard.

- a.** Briefly explain what an installation wizard is. (2)
- b.** Activation codes are usually sent via email. Explain what an activation code is. (2)
- c.** Explain the difference between a full and custom installation. (4)
- d.** The club members sometimes need to remove programs. Outline the correct way of uninstalling software. (2)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	List and describe basic system requirements and how they relate to software?		
2.	Discuss the difference between installing software from an online download and a portable storage medium?		
3.	Describe the installation process?		
4.	Explain how to manage files and folders on a computer?		

CHAPTER
6SOCIAL IMPLICATIONS
OF HARDWARE

CHAPTER OVERVIEW

Unit 6.1	Input and output devices for physically challenged users
Unit 6.2	Health issues due to input and output devices
Unit 6.3	Protecting your hardware
Unit 6.4	Power supply and settings

By the end of this chapter, you will be able to:

- Identify input and output devices for physically challenged users
- Discuss the health issues caused by input and output devices
- Describe how to protect your hardware from theft and damage

INTRODUCTION

This chapter will focus on input and output devices for physically challenged users, the health issues associated with hardware devices, how to protect your devices from theft or damage and how to protect your devices from **power surges** and power failures.

6.1 Input and output devices for physically challenged users

Input and output devices for the physically challenged are used to help people with physical disabilities to interact with the modern world.

INPUT DEVICES

The author of *A Brief History of Time*, famous scientist Stephen Hawking, was diagnosed with a disease called amyotrophic lateral sclerosis (ALS) in 1963 and given only two years to live. He would go on to live for 55 more years (until his death in 2018), but the ALS would cause him to lose the ability to control his muscle movements. An operation done in 1985 to save his life left him unable to speak.

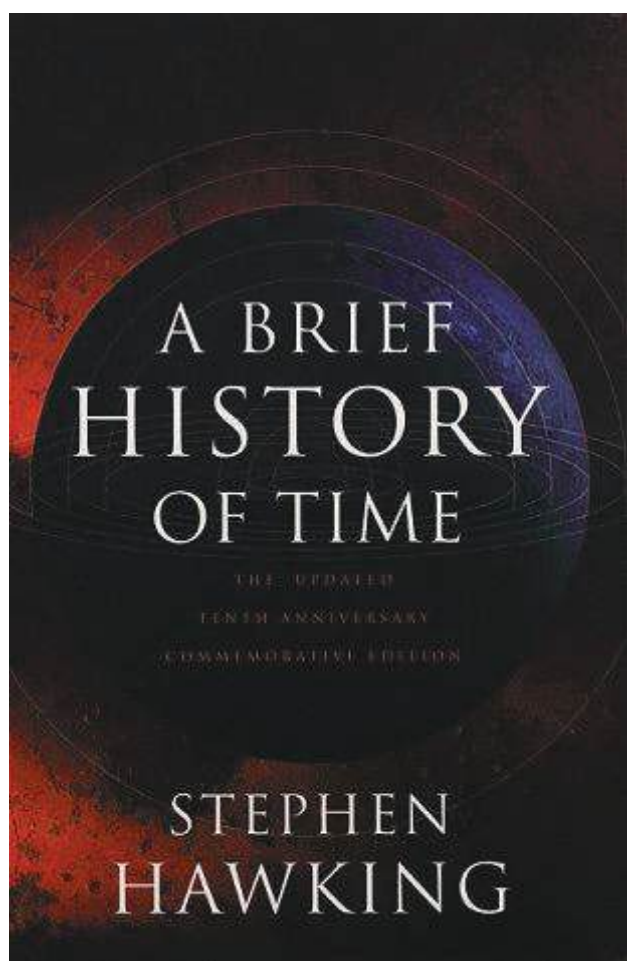


Figure 6.1: *A Brief History of Time* by Stephen Hawking

A Brief History of Time was written using two things, a piece of software called Equalizer that took Professor Hawking's eye movements and translated them into words, and a sensor in his glasses that could track those eye movements. With the help of this technology, a person who would not have been able to use normal computer equipment was able to create ground-breaking work.

There are a variety of alternative input devices available to physically challenged users. These range from fairly common and easy-to-implement adaptations to traditional input devices, to more specialised devices. Next, some of these devices and their advantages and disadvantages are discussed.

INPUT DEVICES FOR PHYSICALLY CHALLENGED USERS

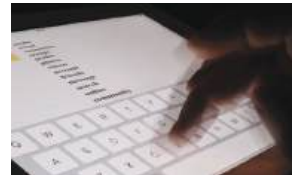
Many options are available to help people who are physically challenged when using ICT. These are just a few devices that could be used.

- **Braille keyboard and Braille display device:** These are keyboards with Braille lettering, that can be used by blind or visually impaired users.
- **On-screen or virtual keyboards:** The keyboard is displayed on a user's screen and used with gesture-controlled pointers or joysticks.
- **Microphones:** Are used with speech recognition software, allowing the user to dictate their input to the computer.
- **Keyguards:** Metal or plastic plates that allow a user to rest his/her hands on a keyboard without accidentally pressing the keys.
- **Head-mounted pointers:** These are controlled by switches to simulate the function of a mouse. The switches can be in hand or foot pads, or pedals or sensors that detect eye movement or facial expressions.
- **Gesture-recognition devices:** The device recognises hand gestures, head or eye movements, or read lips to input information into a computer.
- **Sip-and-puff devices:** A switch that operates the computer when users breathe into (puffing) or through (sipping) special tubes. Special software is used to interpret the sips and puffs and turn them into data that the computer can use.
- **Magnification devices:** Enlarges the information displayed on the computer screen in a range of magnifications and a variety of fonts. The magnifier may have the ability to create a large, scrolling virtual screen, or may only magnify the portion of the display near the mouse pointer.
- **Large-key keyboards:** The print on the number and letter keys of the keyboard are three times larger than those on standard keyboards. The bright white printing on the large black keys increases visibility and the contrast makes it easier to read. These features are helpful to those with vision impairment and for those who wear reading glasses or bifocals, or work in low-lit areas.
- **Braille printers:** These printers operate by embossing raised braille dots onto Braille paper. Pins are pressed into one side of the paper in order to create raised dots on the other side. This is only output.

Keyboard and mouse settings can also be customised to make input easier for physically challenged users. Sticky keys allow users to use common keyboard shortcuts without needing to hold down all the keys at the same time, for example, when using the *Paste* command, users do not need to hold down the *Ctrl* and *V* keys at the same time. They can instead press the keys one at a time.



Braille keyboard



On-screen or virtual keyboards



Microphones



Keyguards



Head-mounted pointers



Gesture-recognition devices



Sip-and-puff devices

You can turn sticky keys on in one of two ways.

1. You can press the *Shift* key five times to turn sticky keys on and off.
2. If this does not work, you can click on the *Windows* key or press the *Windows* key on your keyboard, click on the *Settings* icon and then click on *Ease of Access*. You can then scroll down to the keyboard settings and tick or untick the *Sticky Keys* box.

OUTPUT DEVICES

Physically challenged users can also use a variety of output devices to access information on their computers. Headphones and speakers can, for example, be used with various types of screen-reading software, such as screen-magnifying programs or physical magnifiers mounted onto their screens, to make sections of the screens easier to read.

Braille interfaces work by raising metal or plastic pins on a surface, allowing blind users to read what is on their screens. These devices can only read basic text documents.



Figure 6.2: Computer with Braille display

Braille printers can be used to aid visually impaired users by printing the raised **patterns** of dots on special paper, thereby allowing them to read the text with their fingers.



Activity 6.1

1. Multiple choice:
 - a. Stephen Hawking was diagnosed with?
 - A. ALS
 - B. ADHD
 - C. CTS
 - D. Parkinson's disease
 - b. Which of the following is NOT a device for the physically challenged?
 - A. Microphones
 - B. On-screen or virtual keyboards
 - C. Monitor
 - D. Keyguards

... continued



Activity 6.1

... continued

2. Match Column B with Column A:

COLUMN A	COLUMN B
2.1. A device that recognises hand gestures.	A. Sip-and-puff device
2.2. A device that is used by everybody, not just physically challenged users.	B. Keyguards
2.3. A switch that operates the computer when users breathe into or through special tubes.	C. Gesture-recognition device
2.4. An output device used by physically challenged users.	D. Screen reader
2.5. Metal or plastic plates that allow users to rest their hands on a keyboard without accidentally pressing the keys.	E. Microphone

3. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
- Braille interfaces are for deaf people.
 - The *Paste* keyboard shortcut uses the *Ctrl + P* keys.
 - Sighted users may not be able to use screen readers.
4. List the advantages and disadvantage of the following devices:
- On-screen or virtual keyboards
 - Braille keyboards
 - Head-mounted pointers
 - Gesture-recognition devices
 - Sip-and-puff devices

6.2 Health issues due to input and output devices

With more and more people using computers in their daily lives, certain health issues surrounding computer use have come up in recent years. Some of these have to do with the impact that using a computer has on your body, for example, **repetitive strain injuries (RSIs)** to wrists and back and eye strain from bad posture and lighting.

This section looks at how to prevent these health issues to make sure that you stay healthy.

ERGONOMICS

Ergonomics is concerned with how to design/arrange furniture and equipment in a way that will make people more comfortable and efficient in their work. It also looks at how people interact with their input and output devices to minimise the amount of physical strain placed on their bodies. Therefore, ergonomically designed computer mice, keyboards, desks and chairs have been developed to reduce RSIs, **carpal tunnel syndrome** and back strain.

Ergonomics also determine how people should sit (their posture), how they should look at a screen and how often they should get up.

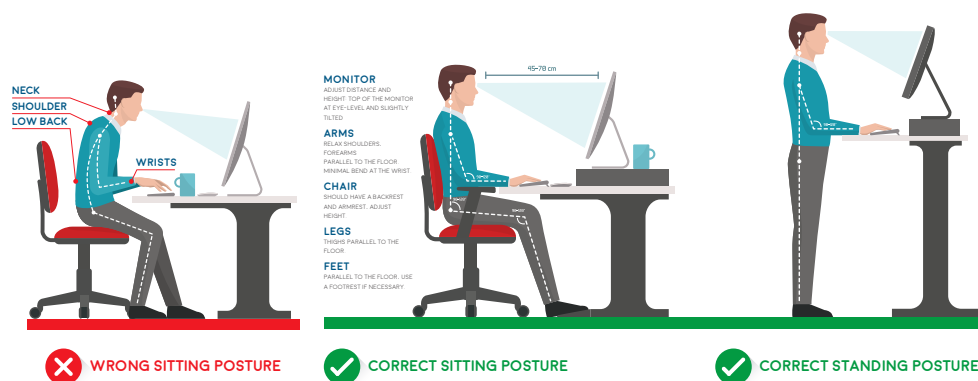


Figure 6.3: Good ergonomics

When you want to reduce the amount of strain on your body, remember to do the following:

- Make sure that your desk, chair and screen are all at a comfortable height.
- Sit properly and do not slouch or hunch your shoulders. This could cause backache and neck pain.
- Stand up and move around at least once an hour. Try using a standing desk (if possible) for some of the time you are working.
- Never bend your neck to look at your screen. Always make sure that your chin is parallel to your desktop.
- Make sure your shoulders are relaxed when you are typing (not raised or hunched), your knees are at a 90° angle and your feet are flat. If you are too short for your feet to touch the floor and have your arms at a comfortable angle, find a footrest to put your feet on.
- Make sure your back and hips are in contact with the back of the chair you are sitting on and that the chair has good back support.

COMPUTER ERGONOMICS

<https://www.youtube.com/watch?v=bLBKUbnLYTs>

Computer and smartphone screens can be set to have less blue light, lessening the glare and reducing eye strain. They can also be tilted until you find a comfortable viewing angle.



Activity 6.2

1. Multiple choice:
 - a. Which of the following equipment does not carry a lot of germs?
 - A. Smartphones
 - B. Computer mice
 - C. Keyboards
 - D. Monitors
 - b. Which one of the following is not an effect of sitting for too long in front of a computer?
 - A. Obesity
 - B. Increased blood pressure
 - C. Hair loss
 - D. High blood sugar
 - c. Which of the following is not an injury due to improper ergonomics?
 - A. Repetitive strain injury
 - B. Computer vision syndrome
 - C. Repetitive strain injury
 - D. Back strain
2. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. Repetitive strain injuries and eye strain are examples of health issues due to input and output devices.
 - b. Hardware needs to be protected from theft and damage.
 - c. Computer and smartphone screens can be set to have less red light, lessening the glare and reducing eye strain.
 - d. You will have good back support when you place your hips and back on the back of the chair.
3. Answer the following questions in your own words:
 - a. Define ergonomics.
 - b. List two ways that you can keep your hardware clean and hygienic.
 - c. List three ways that you can reduce the amount of strain on your body.

6.3 Protecting your hardware

Knowing how to protect your hardware so that it carries on working correctly is as important as knowing what hardware is used for and the health risks associated with using input and output devices. This unit looks at protecting your hardware from theft, damage and power surges. You will also learn about the optimal power settings for your devices and how your devices can contribute to energy savings.

HARDWARE PROTECTION

Hardware needs to be protected from two major things, namely theft and damage. Because your hardware is expensive and sensitive equipment, you must make sure that you do your best to keep it safe and in good working condition for as long as possible.

To protect your hardware from theft, you can do the following:

- Use locking cables attached to walls or hard-to-move desks to lock items such as monitors into place. Use lockable cages or boxes for things such as computer cases and servers.
- Invest in lockable docking stations for laptops. These docks lock as soon as you click a laptop into place, keeping your laptop safe from theft.
- Always be alert and aware of your surroundings when using mobile computing devices such as smartphones and tablets.

To protect your hardware from everyday wear and tear, you should keep the following in mind:

- Always use a screen protector and case for smartphones and tablets.
- Avoid spilling liquids on keyboards and mice.
- Make sure that any device drivers you need are up to date and correctly installed.



Figure 6.4: Do not spill liquid on computing devices



Activity 6.3

1. Answer the following questions in your own words:
 - a. List four ways that you can protect your hardware from theft.
 - b. Give three examples of ways to protect your hardware devices from damage.

6.4 Power supply and settings

Each piece of hardware you use will need to draw on some kind of power to operate, whether it is plugged directly into a power supply (such as monitors and desktop computers), operates with batteries (such as most wireless devices and smartphones, tablets and laptops), or draws power from the computer it is plugged into (such as wired keyboards and mice).

Computers and their associated hardware are some of the largest consumers of power in the home and workplace. By learning how to set these devices so that they use the minimum amount of power needed to operate effectively, and by purchasing devices that have a low energy consumption, you can reduce the amount of power your computer consumes.

Most displays and touchscreens can be adjusted to consume less power. You can do this by adjusting the brightness of your display until it is at a comfortable level. Most screens have a button on the bottom right-hand corner to turn the brightness up or down. Laptops usually have brightness keys on the keyboard and smartphones and tablets can be adjusted by going into the display settings.

You should adjust your screen brightness to suit the area, time of day and lighting. Even lowering the brightness by a small amount can save power and limit the strain you put on your eyes.

When you buy new hardware, you should look for devices that have an **Energy Star rating**. Devices with an Energy Star rating are usually more energy efficient, meaning they use less energy to perform the same tasks as devices that are not rated.

HOW TO PROTECT YOUR DEVICES FROM POWER FAILURES

Uninterruptible power supplies (UPSs) can protect your devices from unexpected power failures and even power surges. UPSs are plugged directly into the main power supply to charge their batteries and will immediately start providing power to the computers connected to them in the case of a power failure.



Activity 6.4

1. Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)
 - a. Devices with an Energy Star rating are usually more energy efficient.
 - b. All the data on your hard drive will be lost in case of a power failure.
 - c. Your display should never be brighter or darker than the area you are using it in.
2. Answer the following questions in your own words:
 - a. Describe a UPS.
 - b. Explain how you would protect your device from power failure.

REVISION ACTIVITY

QUESTION 1: MULTIPLE CHOICE

- 1.1 Which of the following is an output device for physically challenged computer users? (1)
 - A. Gesture-recognition software
 - B. Braille interfaces
 - C. Sip-and-puff devices
 - D. Braille keyboards
- 1.2 What is an advantage of a keyguard? (1)
 - A. It reduces the number of typing errors.
 - B. It does have a universal standard for its keyboard.
 - C. It can be operated with multiple input devices.
 - D. It can be used with speech-recognition software.
- 1.3 Text-to-speech software can be used by which of the following output devices? (1)
 - A. Monitor
 - B. Keyboard
 - C. Printer
 - D. Speaker
- 1.4 Ergonomics does not involve _____. (1)
 - A. Making sure people are in a comfortable position
 - B. Arranging furniture and equipment
 - C. Reducing strain on the human body
 - D. Adjusting the lighting of an area
- 1.5 Which of the following will not reduce your computer's power usage? (1)
 - A. Unplugging your computer when you are not using it
 - B. Reducing the brightness of your display
 - C. Buying hardware with an Energy Star rating
 - D. Reducing your CPU's speed

QUESTION 2: TRUE OR FALSE

Write True or False next to the question number. Correct the statement if it is FALSE. Change the underlined word(s) to make the statement TRUE. (You may not simply use the word NOT to change the statement.)

- a. An eye-motion sensor is activated when a user breathes into its special tubes. (1)
- b. A foot-operated mouse can be used by people who are blind. (1)
- c. Text-to-speech software uses microphones. (1)
- d. The interaction between the human body and computing hardware refers to economics. (1)
- e. Looking at a computer monitor for a long time can damage your sight. (1)

QUESTION 3: MATCHING ITEMS

- 3.1 Choose a term/concept from Column B that matches a description in Column A. Write only the letter next to the question number (e.g. 1J). (5)

COLUMN A	COLUMN B
1. Something that protects your computer from the damage caused by power cuts.	A. Microphone
2. A standard that determines if a device is energy efficient.	B. Screen protector
3. An object that protects a device from scratches and damage.	C. Energy Star rating
4. An object that locks computing devices in place.	D. Wireless keyboard
5. A device that can use biometrics to identify a user.	E. UPS
	F. Padlock
	G. Ergonomics
	H. Braille printer
	I. Virtual keyboard

... continued

REVISION ACTIVITY

... continued

QUESTION 4: SCENARIO-BASED QUESTIONS

Mr Riaad is a primary school teacher who teaches eight and nine-year-olds. He wants to teach them how to use computers safely. Since they will be using the computers a lot, he is also concerned about their eyesight.

- 4.1 Mention two devices that Mr Riaad's learners can use to improve their interaction with computers. Give a reason for each of your answers. (4)
- 4.2 Mention one way his learners may not benefit from using each of these devices. (2)
- 4.3 Mention three things Mr Riaad's learners can do to reduce the strain on their eyes. (3)
- 4.4 One of his learners has been struggling to see the letters on her keyboard. Suggest two alternative input methods that she could use. (2)
- 4.5 A head-mounted pointer is an input device for physically challenged users.
 - a. Explain how a head-mounted pointer is operated. (2)
 - b. Give two disadvantages of using a head-mounted pointer. (2)
- 4.6 Mr Riaad must travel a lot. Name three things he can do to protect his smartphone and laptop while travelling. (3)
- 4.7 Mr Riaad often uses trains to travel from one place to another. During this time, he uses his laptop to complete some of his work. Provide four tips he can use to reduce the strain on his body. (4)
- 4.8 List three ways that Mr Riaad can teach his learners to save energy. (3)

TOTAL: [40]

AT THE END OF THE CHAPTER

NO	CAN YOU ...	YES	NO
1.	Identify input and output devices for physically challenged users?		
2.	Discuss the health issues caused by input and output devices?		
3.	Describe how to protect your hardware from theft and damage?		